Cyber Threat Defense Workbench

Expert briefing | Network-flow attack detection on UNSW-NB15

RESEARCH QUESTION	Can readable detection signals help models rank attack-labeled network flows, and what review workload is needed?

Executive summary

This one-notebook research MVP compares three standalone detection tracks with three alternative classifiers on the public UNSW-NB15 network-flow benchmark. XGBoost ranked attack labels best on the held-out primary test and, at the same alert count as the rule union, found more attack-labeled flows with far fewer normal-labeled alerts. This supports a workload-aware benchmark comparison; it does not establish performance in live enterprise traffic.

73,791	**0.9825**	**27,973**	**99.7%**
primary-test flows	XGBoost test AP	equal alert budget	XGBoost precision

Study design

The published split contains 175,341 training flows and 82,332 test flows. The official CSVs have no timestamps, so a grouped random split within training data provides fit and validation data; it is not a future-period simulation. The primary test excludes 8,541 rows with exact feature overlap against training, leaving 73,791 rows and 38,263 attack labels (51.9%).

The notebook evaluates fixed readable rules, an Isolation Forest novelty score, and a recent-connection concentration score as separate alert tracks. It compares logistic regression, XGBoost, and Explainable Boosting Machine classifiers trained with shared raw inputs and track signals. Validation average precision selected XGBoost before test scoring.

Interpretation guardrail

UNSW-NB15 is cyber-range data with an unusually high attack-label rate. Alert precision and workload figures describe this benchmark. They should not be read as expected live SOC precision, incident counts, or operational savings.

Results and review workload

Ranking performance on the primary test

Classifier	Average precision	ROC AUC	Brier score
Logistic regression	0.9696	0.9654	0.1088
XGBoost (selected)	**0.9825**	**0.9796**	**0.0929**
Explainable Boosting Machine	0.9746	0.9710	0.1049

Average precision summarizes ranking across thresholds; ROC AUC measures ranking across positive and negative pairs. Brier score measures probability error; lower is better.

Equal number of alerts: rules vs XGBoost

Alert source	Alerts	Attack-labeled	Normal-labeled	Precision	Recall
Rules union	27,973	22,578	5,395	80.7%	59.0%
XGBoost, same count	**27,973**	**27,899**	**74**	**99.7%**	**72.9%**

At this benchmark workload, the model queue contains 5,321 more attack-labeled flows and 5,321 fewer normal-labeled flows than the rule queue. Rules still surface 1,626 additional attack-labeled flows outside the equal-size model queue, while adding 5,373 normal-labeled flows. This is a recall-versus-review-volume tradeoff, not an established operating policy.

Three standalone tracks

Track	Alerts	Attack-labeled	Precision	Recall
Rules union	27,973	22,578	80.7%	59.0%
Isolation Forest	5,889	4,039	68.6%	10.6%
Connection concentration	7,157	7,157	100.0%	18.7%

The connection-concentration track produced no normal-labeled alerts in this primary test. Treat this as a dataset-specific pattern, not evidence of a zero-false-alert method in real traffic.

Threshold sensitivity

A cutoff fixed at the validation 90th percentile yielded 1,831 XGBoost test alerts and 4.8% recall. A separate fixed 10% test review budget ranked 7,379 flows and captured 19.3% of attack labels. These answer different questions: a frozen validation cutoff versus a fixed-size ranked queue.

Misses, limits, and practical reading

What the selected model missed

At the rule-matched alert budget, XGBoost missed 10,364 of the 38,263 attack-labeled primary-test flows. Its largest missed categories were Fuzzers (4,747), Exploits (2,607), and Reconnaissance (1,619). The notebook includes global feature importance and a local SHAP explanation for one missed flow. These describe model behavior; they do not establish the cause or intent of an event.

What this experiment can support

• Within this dataset: XGBoost had the strongest test ranking metrics among the compared classifiers, after validation selection.
• At a matched queue size: the selected model captured more attack labels than the fixed rule union in the primary test.
• For investigation: readable rule signals and feature explanations provide examples to inspect, including categories the model misses.

What it cannot establish

Limit	Implication
Base-rate transfer	Attack labels make up 51.9% of the primary test. Precision and false-alert counts will change with the much lower attack prevalence common in operational monitoring.
Future-period performance	The published files have no timestamps. Grouped random validation and held-out testing are not a chronological deployment simulation.
Data duplication and labels	There are 74,301 repeated-input rows in training and 28,386 in published test. 229 training groups share identical inputs but conflicting labels. Deduplication and test-overlap removal reduce one leakage risk; they do not remove all traffic correlation.
Feature availability	Recent-connection counts are supplied dataset fields and cannot be independently reconstructed from the two split CSVs.
Operational readiness	The rules and concentration formula are illustrative. The project has no live sensor, production feature pipeline, case workflow, analyst feedback, or automatic containment.

Suggested next research step

Treat this as a reproducible benchmark MVP. A stronger follow-up would use timestamped traffic with a chronological evaluation split, estimate alert precision at a realistic operational base rate, and verify required features against available telemetry. Those data are outside this project.

Data and source

Dataset: UNSW-NB15, described by Moustafa and Slay, “UNSW-NB15: a comprehensive data set for network intrusion detection systems,” MilCIS 2015, DOI 10.1109/MilCIS.2015.7348942. Dataset documentation: research.unsw.edu.au/projects/unsw-nb15-dataset. Metric values are from the repository notebook and findings document.